

Notebook 02 — Sentiment Analysis

Project: Ad Engagement Analysis

Goal: Analyze emotional tone of user comments on ad campaigns

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from textblob import TextBlob
import warnings
warnings.filterwarnings('ignore')
sns.set_theme(style="whitegrid")

# Load Dataset
try:
    df = pd.read_csv('../data/comments_cleaned.csv')
    print("Dataset 'comments_cleaned.csv' loaded successfully.")
except FileNotFoundError:
    print("Dataset not found error.")

# Check the string comments
display(df[['comment']].head())
```

Dataset 'comments_cleaned.csv' loaded successfully.

	comment
0	unde at dolorem
1	quae ea ducimus
2	alias a voluptatum
3	facere suscipit sunt
4	totam eligendi quaerat

TextBlob Sentiment Calculation

Let's apply TextBlob to get the polarity (-1 to 1) for every comment in the dataset.

```
In [2]: # Create Polarity function
def get_sentiment(text):
    if pd.isna(text):
        return 0.0
    return TextBlob(str(text)).sentiment.polarity

# Calculate Sentiment Polarity
df['sentiment_polarity'] = df['comment'].apply(get_sentiment)

# Categorize based on strict limits
def categorize_sentiment(polarity):
    if polarity > 0.1:
        return 'Positive'
    elif polarity < -0.1:
        return 'Negative'
    else:
        return 'Neutral'

df['sentiment_category'] = df['sentiment_polarity'].apply(categorize_sentiment)

print("Sentiment Distribution:")
print(df['sentiment_category'].value_counts())
```

Sentiment Distribution:
sentiment_category
Neutral 7488
Name: count, dtype: int64

Sentiment Distribution Plots

```
In [3]: # Plot Density Distribution of Polarity
plt.figure(figsize=(10,6))
sns.histplot(df['sentiment_polarity'], bins=30, kde=True, color='teal')
plt.title('Distribution of Comment Sentiment Polarity')
plt.xlabel('Polarity (-1 Negative to 1 Positive)')
plt.ylabel('Density')
plt.show()

# Plot categorical Sentiment
plt.figure(figsize=(8,6))
sns.countplot(x='sentiment_category', data=df, palette=['grey', 'green', 'red'])
plt.title('Gross Count of Sentiment Categories')
plt.show()
```



